College of Computer and Information Sciences
Department of Information Systems

IS 220 – Database Fundamentals
Project

First Semester 2025
Phase #3



Wsslini database system

Phase # 3

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Introduction

The Wasslini System Project is a database system designed to represent a ride-booking application similar to modern smart transportation platforms.

The project aims to organize and store data related to customers, drivers, trips, payments, and ratings using the Entity-Relationship Diagram (ERD) model.

The system connects customers with drivers to complete trips and record all related details, while also managing payment and rating processes after each trip to ensure service quality and enhance the overall user experience.

System Requirements:

The wsslini system works as a platform that manages customers drivers and customer service, customers submit trip request that include pick up and drop off locations. Available drivers each assigned to a specific vehicle, can accept these requests. When a trip starts, details, such as time, distance, and fare are recorded. And after completing the trip, the customer makes the payment through the system and payment details are saved, and both the customer and driver rate each other, which updates their rating averages.If any problem occurs, the customer can open a support ticket which is handled by customer service according to their shift schedule.

User Identification and Tasks

The following table1 identifies the main users of the Wsslini database system and outlines the key tasks each user is expected to perform. These roles interact with the system to manage trips, payments, and overall service quality.

Users	Activities or Tasks
Customers	• Register/Login into the system • Request a trip by setting pickup and destination locations • View trip status and assigned driver/vehicle • Make the payment after trip completion • Give rating and comments for the trip
Driver	• Register/Login into the system • View and accept trip requests • Start and end assigned trips • Update trip status (Available/Busy)
Customer Support	• Handle customer inquiries and complaints • Monitor trip details and payment status to resolve issues

Table1. Users of Wsslini database system and their tasks

Entity Relationship Diagram (ERD):

Figure 1 illustrates the Entity-Relationship Diagram (ERD) for the wsslini database system. It provides a clear overview of the main entities and how they are connected within the database.

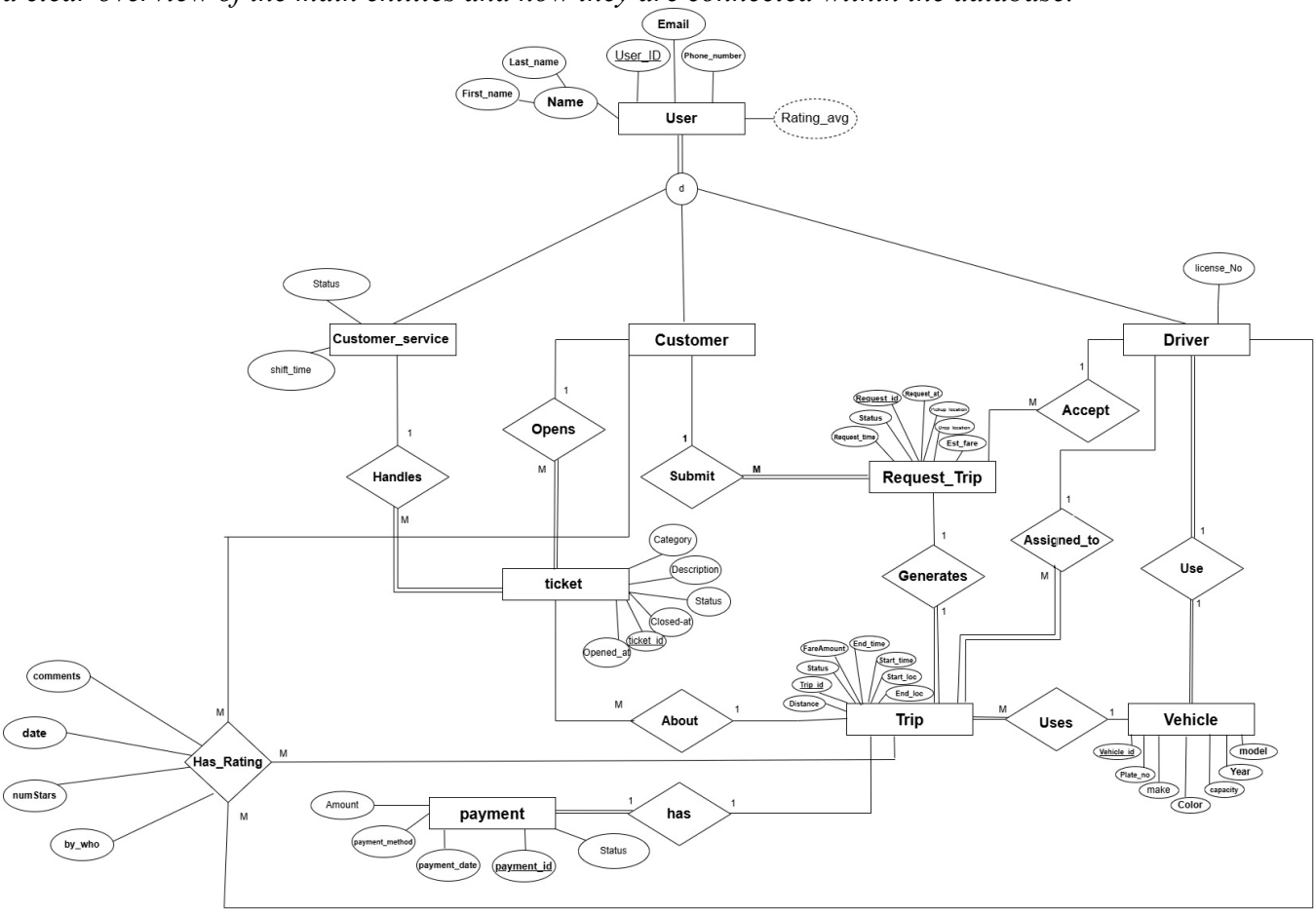


Figure 1. ERD of Wsslini database system

Mapping Entity Relationship Diagram (ERD) To Relational Schema :

Custmor (User_id, Email , Phone_number, First_name, Last_name)

Driver (User_id , Email , Phone_number , First_name , Last_name , license_No)

Customer_service (User_id, Email,Phone_number, First_name, Last_name,shift_time, status)

Request_Trip (Request_id , Status , Request_time , Est_fare , Pickup_location , Drop_location , Request_at , Customer_id , Driver_id)

FK: Customer_id references to Customer(User_id)

FK: Driver_id reference to Driver(User_id)

Trip (Trip_id , Distance , Status , FareAmount , End_time , Start_time , Start_loc , End_loc , Driver_id , Request_id , Vehicle_id)

FK: Driver_id references to Driver(User_id)

FK: Request_id references to Request_Trip(Request_id)

FK: Vehicle_id references to Vehicle(Vehicle_id)

Ticket (ticket_id, Trip_id, Customer_id, Customer_Service_id, category, Description, status,Closed_at, Opened_at)

FK: Trip_id reference Trip(trip_id)

FK: Customer_id reference Customer(user_id)

FK: Customer_Service_id reference Customer_service(user_id)

Has_Rating (Customer_id , Driver_Iid , trip_id , comments, date , numStars , by_who)

FK: Customer_id reference Customer(User_id)

FK: Driver_id reference Driver(User_id)

FK: trip_id reference Trip (trip_id)

Vehicle (model , Year , Capacity , Color , Make , plate_on , Vehicle_id , User_id)

Fk: User_id reference Driver(User_id)

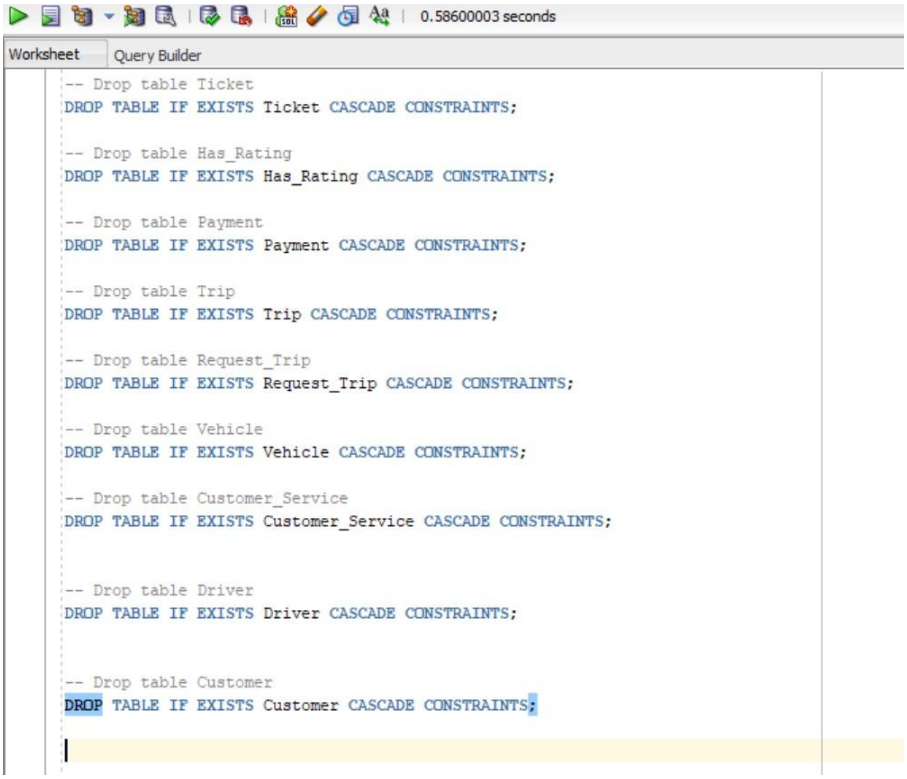
Payment (Payment_id , Status , payment_date , payment_method , Amount , Trip_id)

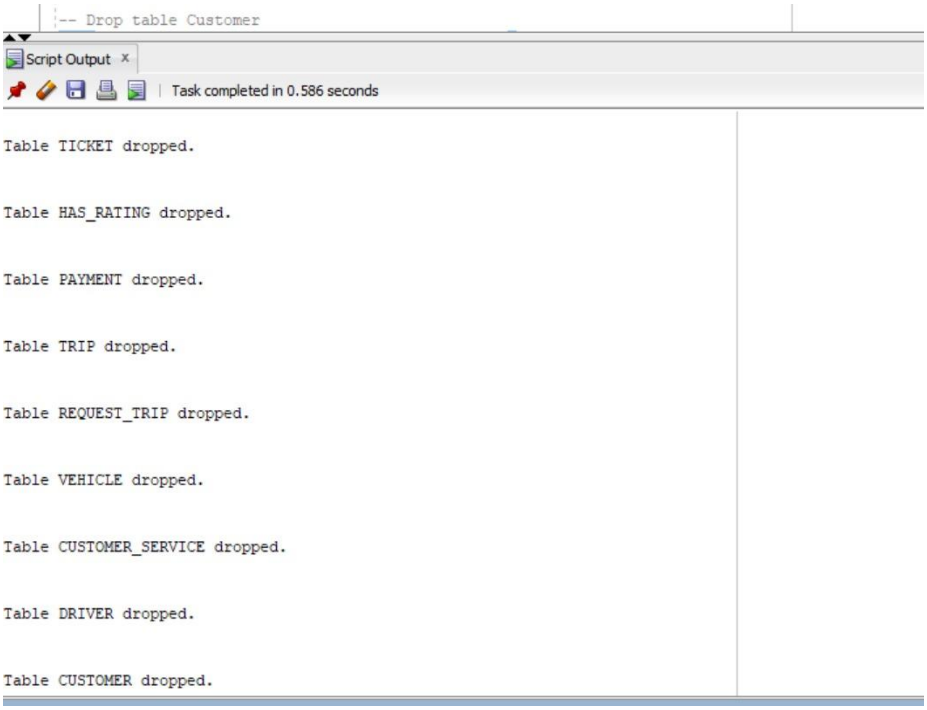
FK: Trip_id reference Trip(trip_id)

Implementation phase

Submit the COMMANDS and the resulted OUTPUT (snapshot) from the execution of the DB table's creation, data insertion and the data queries.

1. Create Database Tables





```
CREATE TABLE Customer (
    User_id NUMBER PRIMARY KEY,
    Email VARCHAR2(50) NOT NULL UNIQUE,
    Phone_number VARCHAR2(15)  NOT NULL,
    First_name VARCHAR2(10)  NOT NULL,
    Last_name VARCHAR2(10)  NOT NULL
);

CREATE TABLE Driver (
    User_id NUMBER PRIMARY KEY,
    Email VARCHAR2(50) UNIQUE,
    Phone_number VARCHAR2(15),
    First_name VARCHAR2(10),
    Last_name VARCHAR2(10),
    License_No VARCHAR2(15)
);

CREATE TABLE Customer_Service (
    User_id NUMBER PRIMARY KEY,
    Email VARCHAR2(50) UNIQUE,
    Phone_number VARCHAR2(15),
    First_name VARCHAR2(10),
    Last_name VARCHAR2(10),
    Shift_time VARCHAR2(10),
    Status VARCHAR2(20)
);

CREATE TABLE Vehicle (
    Vehicle_id NUMBER PRIMARY KEY,
    Model VARCHAR2(50) NOT NULL,
    Year NUMBER NOT NULL,
    Capacity NUMBER NOT NULL,
    Color VARCHAR2(10),
    Make VARCHAR2(20),
    Plate_on VARCHAR2(20) UNIQUE,
    User_id NUMBER Not NULL,
  CONSTRAINT  fk_vehicle_driver FOREIGN KEY (User_id) REFERENCES Driver(User_id)
);
```

```
CREATE TABLE Request_Trip(
  Request_id NUMBER PRIMARY KEY,
  Status VARCHAR2(20),
  Request_at DATE,
  Est_fare DECIMAL(10,2),
  Pickup_location VARCHAR2(100),
  Drop_location VARCHAR2(100),
  Customer_id NUMBER NOT NULL,
  Driver_id NUMBER NOT NULL,
  CONSTRAINT fk_request_customer FOREIGN KEY(Customer_id) REFERENCES Customer(User_id),
    CONSTRAINT fk_request_driver FOREIGN KEY(Driver_id) REFERENCES Driver(User_id)
);
```

```
CREATE TABLE Trip (
    Trip_id NUMBER PRIMARY KEY,
    Distance NUMBER,
    Status VARCHAR2(10),
    FareAmount NUMBER,
    End_time TIMESTAMP,
    Start_time TIMESTAMP,
    Start_loc VARCHAR2(50),
    End_loc VARCHAR2(200),
    Driver_id NUMBER NOT NULL,
    Request_id NUMBER NOT NULL,
    Vehicle_id NUMBER NOT NULL,
    CONSTRAINT fk_trip_driver FOREIGN KEY (Driver_id) REFERENCES Driver(User_id),
    CONSTRAINT fk_trip_request FOREIGN KEY (Request_id) REFERENCES Request_Trip(Request_id),
    CONSTRAINT fk_trip_vehicle FOREIGN KEY (Vehicle_id) REFERENCES Vehicle(Vehicle_id)
);
```

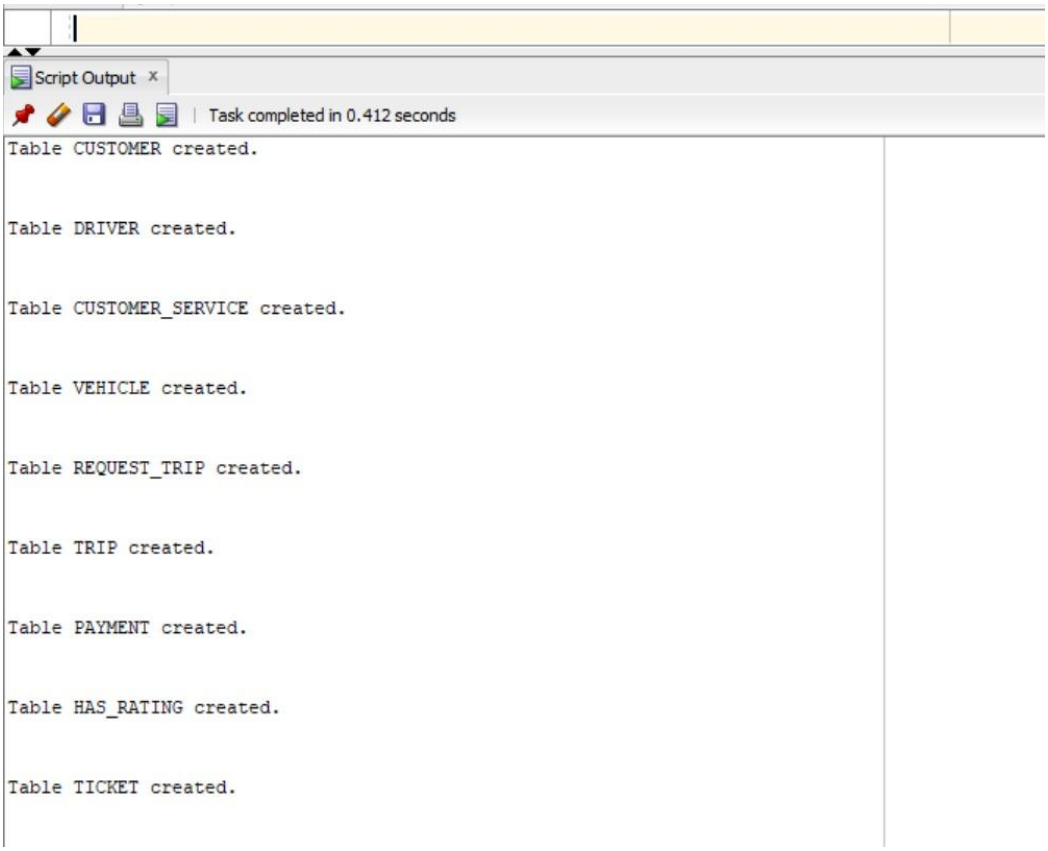
```
CREATE TABLE Payment (

Payment_id NUMBER PRIMARY KEY,
Status VARCHAR2(15),
Payment_method VARCHAR2(20),
Amount NUMBER,
Trip_id NUMBER NOT NULL,
CONSTRAINT fk_payment_trip FOREIGN KEY(Trip_id) REFERENCES Trip(Trip_id)

);
```

```
CREATE TABLE Has_Rating (
    Customer_id NUMBER NOT NULL,
    Driver_id NUMBER NOT NULL,
    trip_id NUMBER NOT NULL,
    comments VARCHAR2(200),
    numStars NUMBER(1) CHECK (numStars BETWEEN 1 AND 5),
    by_who VARCHAR2(10) CHECK (by_who IN ('customer', 'driver')),
    CONSTRAINT pk_rating PRIMARY KEY (Customer_id, Driver_id, trip_id),
    CONSTRAINT fk_rating_customer FOREIGN KEY (Customer_id) REFERENCES Customer(User_id),
    CONSTRAINT fk_rating_driver FOREIGN KEY (Driver_id) REFERENCES Driver(User_id),
    CONSTRAINT fk_rating_trip FOREIGN KEY (trip_id) REFERENCES Trip(Trip_id)
);
```

```
CREATE TABLE Ticket (
    ticket_id NUMBER PRIMARY KEY,
    Trip_id NUMBER NOT NULL,
    Customer_id NUMBER NOT NULL,
    Customer_Service_id NUMBER NOT NULL,
    category VARCHAR2(50),
    Description VARCHAR2(500),
    status VARCHAR2(15),
    CONSTRAINT fk_ticket_trip FOREIGN KEY (Trip_id) REFERENCES Trip(Trip_id),
    CONSTRAINT fk_ticket_customer FOREIGN KEY (Customer_id) REFERENCES Customer(User_id),
    CONSTRAINT fk_ticket_service FOREIGN KEY (Customer_Service_id) REFERENCES Customer_service(User_id)
);
```


Script Output
Task completed in 0.412 seconds
Table CUSTOMER created.
Table DRIVER created.
Table CUSTOMER_SERVICE created.
Table VEHICLE created.
Table REQUEST_TRIP created.
Table TRIP created.
Table PAYMENT created.
Table HAS_RATING created.
Table TICKET created.

2. Insert Data

These are the tables after importing data from excel files:

```
select * from ticket;
commit;
```

Query Result | All Rows Fetched: 5 in 0.003 seconds

	TICKET_ID	TRIP_ID	CUSTOMER_ID	CUSTOMER_SERVICE_ID	CATEGORY	DESCRIPTION	STATUS
1	1122	2001	1	201	payment issue	customer was charge twice for the same trip	resolved
2	2233	2002	2	202	driver behavior	driver was rude and took a longer route	open
3	3344	2003	3	203	lost item	customer forget phone in the car	in progress
4	4455	2004	4	204	technical issues	app showing incorrect payment	resolved
5	5566	2005	5	205	safety concern	driver was using phone while driving	open

```
select * from payment;
commit;
```

Query Result | Script Output | All Rows Fetched: 5 in 0.001 seconds

	PAYMENT_ID	STATUS	PAYMENT_METHOD	AMOUNT	TRIP_ID
1	301	Paid	Card	25	2001
2	302	Paid	Cash	18	2002
3	303	Not Paid	Card	30	2003
4	304	Paid	Cash	22	2004
5	305	Paid	Card	20	2005

```
select * from has_rating;
commit;
```

Script Output | Query Result | All Rows Fetched: 5 in 0.001 seconds

	CUSTOMER_ID	DRIVER_ID	TRIP_ID	COMMENTS	NUMSTARS	BY_WHO
1	1	101	2001	Very good	5	customer
2	2	102	2002	not bad	3	customer
3	3	103	2003	nice	5	customer
4	4	104	2004	good	4	driver
5	5	105	2005	I like the trip	5	customer

```
select * from trip;
commit;
```

Script Output | Query Result | All Rows Fetched: 5 in 0.002 seconds

	TRIP_ID	DISTANCE	STATUS	FAREAMOUNT	END_TIME	START_TIME	START_LOC	END_LOC	DRIVER_ID	REQUEST_ID	VEHICLE_ID
1	2001	60	Complated	60	10-NOV-25 12.00.00.000000000 AM	10-NOV-25 12.00.00.000000000 AM	King Abdullah Road	King Saud University	101	11111	500
2	2002	23	InProgress	20	11-NOV-25 12.00.00.000000000 AM	11-NOV-25 12.00.00.000000000 AM	Olaya Street	Airport	102	22222	501
3	2003	44	Cancelled	33	12-NOV-25 12.00.00.000000000 AM	12-NOV-25 12.00.00.000000000 AM	Airport	Olaya Street	103	33333	502
4	2004	77	Complated	80	13-NOV-25 12.00.00.000000000 AM	13-NOV-25 12.00.00.000000000 AM	Malqa Distric	King Abdullah Road	104	44444	503
5	2005	9	InProgress	10	13-NOV-25 12.00.00.000000000 AM	13-NOV-25 12.00.00.000000000 AM	King Fahad Road	Malqa Distric	105	55555	504

```
select * from request_trip;
```

Script Output | Query Result | SQL | All Rows Fetched: 5 in 0.001 seconds

	REQUEST_ID	STATUS	REQUEST_AT	EST_FARE	PICKUP_LOCATION	DROP_LOCATION	CUSTOMER_ID	DRIVER_ID
1	11111	completed	10-NOV-25	15.13	olaya street	Al Nakheel Mall	1	101
2	22222	pending	12-NOV-25	23.7	King Khalid Airport	Al Hamra Metro Station	2	102
3	33333	accepted	10-NOV-25	5.4	Specialized Hospital	Pharmacy of Medicine	3	103
4	44444	completed	13-NOV-25	18	Cake bakery	Saud Al Kabir Street	4	104
5	55555	accepted	14-NOV-25	37.28	Princess Nourah University	King Fahd Library	5	105

```
select * from vehicle;
commit;
```

Query Result 2 | SQL | All Rows Fetched: 5 in 0.002 seconds

	VEHICLE_ID	MODEL	YEAR	CAPACITY	COLOR	MAKE	PLATE_ON	USER_ID
1	500	Camry	2020	4	White	Toyota	ABC111	101
2	501	Sonata	2021	4	Black	Hyundai	ABD222	102
3	502	Corolla	2019	4	Blue	Toyota	ABC333	103
4	503	Altima	2022	4	Grey	Nissan	XYZ444	104
5	504	Kia Rio	2020	4	Red	KIA	XYZ555	105

```
select * from customer;
commit;
```

Query Result | SQL | All Rows Fetched: 5 in 0.001 seconds

	USER_ID	EMAIL	PHONE_NUMBER	FIRST_NAME	LAST_NAME
1	1	ahmed.saleh@gmail.com	551234567	Ahmed	Saleh
2	2	sara.salem@gmail.com	559876543	Sara	Salem
3	3	noura.khalid@gmail.com	561122334	Noura	Khalid
4	4	yousef.ali@gmail.com	537788991	Yousef	Ali
5	5	rana.fahad@gmail.com	596655442	Rana	Fahad

```
select * from customer_service;
commit;
```

Query Result 1

SQL | All Rows Fetched: 5 in 0.002 seconds

	USER_ID	EMAIL	PHONE_NUMBER	FIRST_NAME	LAST_NAME	SHIFT_TIME	STATUS
1	201	cs.lama@gmail.com	581122113	Lama	Saleh	Morning	Active
2	202	cs.maha@gmail.com	599911211	Maha	Ali	Evening	Active
3	203	cs.fatimah@gmail.com	552211998	Fatimah	Nasser	Night	Inactive
4	204	cs.reema@gmail.com	536611227	Reema	Omar	Morning	Active
5	205	cs.hanin@gmail.com	547812233	Hanin	Saad	Night	Active

```
commit;
select * from driver;
commit;
```

Query Result | Script Output

SQL | All Rows Fetched: 5 in 0.009 seconds

	USER_ID	EMAIL	PHONE_NUMBER	FIRST_NAME	LAST_NAME	LICENSE_NO
1	101	khalid.driver@gmail.com	552244668	Khalid	Ahmed	DRV2024901
2	102	mohammed.driver@gmail.com	535544332	Mohammed	Saad	DRV2024810
3	103	abdullah.driver@gmail.com	507788123	Abdullah	Mohammed	DRV2024999
4	104	fahad.driver@gmail.com	549911223	Fahad	Fawaz	DRV2024775
5	105	saad.driver@gmail.com	568822001	Saad	Waled	DRV2024567

3. Write and Execute SQL Queries

- Query 1: uses WHERE + BETWEEN
- A customer service employee requested this query to see trips with fares between 20 and 30 for checking a customer`s question.

```
---- 1. WHERE + BETWEEN: trips costing between 20 and 30
SELECT Trip_id, FareAmount, Distance, Status
FROM Trip
WHERE FareAmount BETWEEN 20 AND 30;
```

Query Result

SQL | All Rows Fetched: 1 in 0.003 seconds

	TRIP_ID	FAREAMOUNT	DISTANCE	STATUS
1	2002	20	23	InProgress

- Query 2: a nested query and uses aggregate function (MAX)
- A customer service employee requested this to see which customer gave the highest rating.

```
---display the Customer id and the number of stars that has the highesr number
Select Customer_id,numStars from has_rating WHERE numStars = (SELECT MAX (numStars)from has_rating);
```

Query Result | Query Result 1

SQL | All Rows Fetched: 3 in 0.002 seconds

	CUSTOMER_ID	NUMSTARS
1	1	5
2	3	5
3	5	5

- Query 3: uses UNION
- A customer service employee requested this query to view all users together in one report.

```
--  UNION: list all users from customors and drivers
SELECT First_name, Last_name, Email, 'Customer' AS User_Type FROM Customer
UNION
SELECT First_name, Last_name, Email, 'Driver' AS User_Type FROM Driver;
```

Query Result | Query Result 1

SQL | All Rows Fetched: 10 in 0.004 seconds

	FIRST_NAME	LAST_NAME	EMAIL	USER_TYPE
1	Ahmed	Saleh	ahmed.saleh@gmail.com	Customer
2	Sara	Salem	sara.salem@gmail.com	Customer
3	Noura	Khalid	noura.khalid@gmail.com	Customer
4	Yousef	Ali	yousef.ali@gmail.com	Customer
5	Rana	Fahad	rana.fahad@gmail.com	Customer
6	Khalid	Ahmed	khalid.driver@gmail.com	Driver
7	Mohammed	Saad	mohammed.driver@gmail.com	Driver
8	Abdullah	Mohammed	abdullah.driver@gmail.com	Driver
9	Fahad	Fawaz	fahad.driver@gmail.com	Driver
10	Saad	Waled	saad.driver@gmail.com	Driver

- Query 4: uses group by+having
- A customer service employee requested this to see the shift time of all employees who are active

```
-- Shows the shift_time of customer service employees who are Active
SELECT shift_time, status
FROM customer_service
GROUP BY shift_time, status
HAVING status = 'Active';
```

Query Result | Query Result 1 | Query Result 2

SQL | All Rows Fetched: 3 in 0.001 seconds

	SHIFT_TIME	STATUS
1	Evening	Active
2	Morning	Active
3	Night	Active

- Query 5: uses GROUP BY
- A customer service employee requested this query to see how many vehicles belong to each car brand.

```
-- Shows number of vehicles for each brand (GROUP BY only)
SELECT Make, COUNT(*) AS Number_of_Vehicles
FROM Vehicle
GROUP BY Make;
COMMIT;
```

Script Output

Task completed in 0.023 seconds

```
MAKE                 NUMBER_OF_VEHICLES
-------------------- ------------------
Toyota                                2
Hyundai                               1
Nissan                                1
KIA                                   1


Commit complete.
```

- Query 6: uses JOIN
- A customer service employee requested this query to view drivers and the vehicle they use.

```
-- Displays drivers and the vehicles they use
SELECT
    Driver.user_id,
    Driver.first_name,
    Vehicle.vehicle_id,
    Vehicle.model,
    Vehicle.plate_on
FROM Driver
JOIN Vehicle
    ON Driver.user_id = Vehicle.user_id;
```

Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 5 in 0.002 seconds

	USER_ID	FIRST_NAME	VEHICLE_ID	MODEL	PLATE_ON
1	101	Khalid	500	Camry	ABC111
2	102	Mohammed	501	Sonata	ABD222
3	103	Abdullah	502	Corolla	ABC333
4	104	Fahad	503	Altima	XYZ444
5	105	Saad	504	Kia Rio	XYZ555

- Query 7: uses UPDATE
- A customer service employee requested this query to change an employee`s shift time.

```
-- Updates the shift_time to 'Evening' for the employee with User_id = 201
UPDATE Customer_Service
SET Shift_time = 'Evening'
WHERE User_id = 201;
COMMIT;
```

Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Script Output

Task completed in 0.025 seconds

```
1 row updated.


Commit complete.
```

- Customer service table before updating:

Query Result 1

SQL | All Rows Fetched: 5 in 0.002 seconds

	USER_ID	EMAIL	PHONE_NUMBER	FIRST_NAME	LAST_NAME	SHIFT_TIME	STATUS
1	201	cs.lama@gmail.com	581122113	Lama	Saleh	Morning	Active
2	202	cs.maha@gmail.com	599911211	Maha	Ali	Evening	Active
3	203	cs.fatimah@gmail.com	552211998	Fatimah	Nasser	Night	Inactive
4	204	cs.reema@gmail.com	536611227	Reema	Omar	Morning	Active
5	205	cs.hanin@gmail.com	547812233	Hanin	Saad	Night	Active

- Customer service table after updating:

Query Result

SQL | All Rows Fetched: 5 in 0.003 seconds

	USER_ID	EMAIL	PHONE_NUMBER	FIRST_NAME	LAST_NAME	SHIFT_TIME	STATUS
1	201	cs.lama@gmail.com	581122113	Lama	Saleh	Evening	Active
2	202	cs.maha@gmail.com	599911211	Maha	Ali	Evening	Active
3	203	cs.fatimah@gmail.com	552211998	Fatimah	Nasser	Night	Inactive
4	204	cs.reema@gmail.com	536611227	Reema	Omar	Morning	Active
5	205	cs.hanin@gmail.com	547812233	Hanin	Saad	Night	Active

- Query 8: uses DELETE
- A customer service employee requested this to delete a specific rating.

```
-- Deletes the Has_Rating row with trip_id = 2003
DELETE FROM Has_Rating
WHERE trip_id = 2003;

COMMIT;
```

Script Output

Task completed in 0.109 seconds

```
1 row deleted.


Commit complete.
```

- The Has_Rating table before delete:

```
select * from has_rating;
commit;
```

Script Output | Query Result

SQL | All Rows Fetched: 5 in 0.001 seconds

	CUSTOMER_ID	DRIVER_ID	TRIP_ID	COMMENTS	NUMSTARS	BY_WHO
1	1	101	2001	Very good	5	customer
2	2	102	2002	not bad	3	customer
3	3	103	2003	nice	5	customer
4	4	104	2004	good	4	driver
5	5	105	2005	I like the trip	5	customer

- The Has_Rating table after delete:

Query Result

SQL | All Rows Fetched: 4 in 0.002 seconds

	CUSTOMER_ID	DRIVER_ID	TRIP_ID	COMMENTS	NUMSTARS	BY_WHO
1	1	101	2001	Very good	5	customer
2	2	102	2002	not bad	3	customer
3	4	104	2004	good	4	driver
4	5	105	2005	I like the trip	5	customer